

Sanwal Bilal

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SUMMARY

AI/Machine Learning Engineer specializing in the development and deployment of intelligent systems using Python and modern machine learning frameworks. Proficient in cloud computing environments including AWS, Azure, GCP, and Databricks, with experience in building scalable, data-driven solutions. Strong expertise in Spark, deep learning, and AI pipelines, combined with a proactive mindset and a high level of accountability for delivering reliable and efficient solutions.

TECHNICAL SKILLS

Programming & Core Concepts: Python (3.x), OOP, Data Structures & Algorithms, SOLID, Dependency Injection, Design Patterns

AI / Machine Learning: NLP, Computer Vision, Deep Learning, Transformer Models, LLMs (fine-tuning, deployment, integration), Model Evaluation

Frameworks & Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face Transformers

Backend & APIs: Flask, Django, REST API Development

Cloud & DevOps: AWS (Lambda, SQS, SNS, S3, EC2), Azure, GCP, Databricks, Docker, Serverless Architectures

Data Engineering & Databases: SQL, Relational Databases, SQLAlchemy (ORM), Distributed & Eventually Consistent Systems

System Architecture: Microservices, Message Brokers, Event-Driven Systems

Tools: Git, Linux, Data Visualization, System Monitoring & Alerting

EXPERIENCE

Junior Machine Learning Engineer

AI Soft (Pvt) Ltd

Feb 2023 – Present

Islamabad, Pakistan

- Developed, deployed, and maintained scalable AI/ML applications using Python, TensorFlow, and PyTorch, improving system performance by **30%** and reducing processing time by **25%**.
- Designed and deployed ML models for AI-driven features, achieving **90%+** accuracy across key use cases and enhancing user engagement by **20%**.
- Integrated third-party AI/ML APIs into existing web applications, reducing development time by **40%** and improving feature delivery speed.
- Implemented data-driven optimization strategies, increasing model efficiency by **25%** and reducing error rates by **15%**.
- Built and optimized NLP pipelines using open-source LLMs for summarization and information extraction, improving response relevance by **35%**.
- Led NLP and computer vision initiatives, improving team productivity by **20%** through collaboration and mentorship.

PROJECTS

AI-Powered PPE Detection & Reporting System

Computer Vision + LLM Integration

github.com/sanwalbilal/PPE-Detection

- Architected a scalable AI pipeline combining **YOLO-based computer vision** and **LLM-based reporting** for intelligent safety monitoring.
- Engineered real-time inference workflows with optimized frame processing and detection pipelines.
- Developed automated incident reporting using LLMs, reducing manual reporting effort significantly.
- Designed modular micro-pipeline enabling seamless integration with edge hardware and cloud systems.

Real-Time Missing Person Detection with RAG-Based Alerting

Face Recognition + Vector Search (Safe City)

github.com/sanwalbilal/Missing-Person-Detection

- Built end-to-end pipeline: Video Stream → Face Detection → Embedding Generation → Vector Search → Identity Matching.
- Leveraged InsightFace (buffalo.l) for high-accuracy facial embeddings; implemented **RAG pipeline** for large-scale record matching.

- Designed automated alert system with multi-channel notifications (SMS & Email) for real-time incident reporting.
- Achieved ~**90%** recognition accuracy with minimized false accept/reject rates and near real-time detection-to-alert latency.

Autonomous AI Reconnaissance System

LLMs + LangGraph (OSINT)

github.com/sanwalbilal/AI-Recon-System

- Architected an **LLM-powered autonomous agent system** for OSINT-driven reconnaissance and intelligence gathering.
- Engineered dynamic tool orchestration using **LangGraph**, enabling context-aware decision loops.
- Built scalable multi-tool integration pipeline for real-time data enrichment and structured insight generation.
- Designed extensible framework supporting cybersecurity research and automated intelligence workflows.

EDUCATION

University of Mianwali

B.S. in Computer Science – GPA: 3.36/4.0

Mianwali, Pakistan